A Comparative Study on Agentic Frameworks for Fairness in Indian Socio-cultural Contexts

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Today's LLMs are trained on Western-centric data, which can lead to embedded biases when the models are deployed in culturally diverse contexts. This is especially a critical issue in the Indian socio-cultural setting which consists of several planes such as caste, religion, region and language that intersect with each other, resulting in potential stereotyping and misrepresentation. Hence, there is a growing need to study and evaluate the fairness of LLMs in such contexts.

Recent work has revolved around the introduction of agentic frameworks – such as multi-agent deliberation – to improve cultural alignment and sensitivity. In parallel, the last 2 years have also seen the release of India-centric fairness benchmarks that aim to evaluate fairness of LLMs in Indian socio-cultural contexts.

In this term paper, I propose to bring these 2 lines of work together and evaluate whether agentic frameworks can improve fairness, safety and alignment by testing them on Indian socio-cultural scenarios with bias and stereotyping risks.

The aim of this paper is to benchmark multilingual open-weights models using **(1) multi-agent debate**, where multiple agents independently reason and aggregate responses to reduce biased outputs (Ki et al., 2025), and **(2) MACD**, a training-free multi-agent cultural debate framework that assigns culturally distinct personas and synthesizes a balanced response (Tan et al., 2026). Evaluation will be conducted on the **FairI Tales** (Nawale et al., 2025) and **IndiCASA** (Santhosh et al., 2025) datasets. This can also be extended to explore comparisons between observed bias and stereotypes in English prompts vs regional-script prompts.

References

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